

Requirement Analysis Document

Project: Smart Student Assignment & Code Reviewer

Group - 11

Batch - 8B

1. Business Problem & Goal:-

- **The Problem:** The current process of reviewing student code is manual and highly time-consuming for computer science faculty. Students must wait days for basic syntax and logic feedback, which slows down their learning process.
- **The Business Goal:** Build a centralized, automated system that allows students to submit code for instant AI-driven review, while providing teachers with a dashboard to monitor all submissions and feedback.

2. Stakeholders:-

- **Students:** Need instant, constructive feedback to improve their coding skills and complete assignments efficiently.
- **Teachers / Faculty:** Need to efficiently track student submissions and reduce the time spent manually grading basic syntax errors.

3. Actors:-

1. Student
2. Teacher

4. Functional Requirements:-

- **Student:**
 - Register and log in
 - View assigned problems
 - Submit code via an integrated editor
 - Receive AI feedback via email.
- **Teacher:**
 - Register and log in
 - View a dashboard of all student submissions
 - View the code submitted by students
 - View the AI-generated feedback status.

5. Non-Functional Requirements:-

- **Security:**
 - Only authorized users can access the dashboards
 - Students can only view their own feedback, while teachers can view all submissions.
- **Performance / Response Time:** The system should securely trigger the automated AI review workflow without freezing or crashing the student's web browser.
- **Reliability:** The system must accurately route the correct feedback to the correct student's email address.

6. User Stories:-

- **Student:** As a Student, I want to submit my assignment code through a web portal so that I can receive instant, automated feedback on my logic and syntax.
- **Teacher:** As a Teacher, I want to view a centralized dashboard of all submissions so that I can track which students have completed the assignment and see the AI's initial evaluation.

7. Acceptance Criteria:-

User Story: As a Student, I want to submit my assignment code through a web portal so that I can receive instant, automated feedback.

- **Given** the student is logged into the system
- **When** the student pastes valid code into the editor and clicks submit
- **Then** the system should save the code to the database and trigger the automation webhook
- **And** the initial AI Review status should be marked as "Pending" on the teacher's dashboard.

User Story: As a Teacher, I want to view a centralized dashboard of all submissions so that I can track which students have completed the assignment and see the AI's initial evaluation.

- **Given** the teacher is successfully logged into the application
- **When** the teacher navigates to the Teacher Dashboard
- **Then** the system should display a complete, tabular list of all student code submissions
- **And** each entry should clearly show the student's details and whether the AI review is pending or complete.